"Advancing Biodiversity Conservation: Comparative Evaluation of Machine Learning Models for Species Classification"

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Overview

1) **Biodiversity Classification and Nate's RAG Project**

A brief summary of Biodiversity Classification, Nate's RAG Project, and the TOL10M+ and iNaturalist datasets

2) **Models and the Training Process**

An Overview of Hiera and CLIP. How did I set up the training process? What did that look like?

3) **Results + Lessons Learned**

What was the outcome of the evaluations? What lessons did I learn from these datesets and models?

4) **Future RAG methologies, INAT discovery and discussions, Impact**

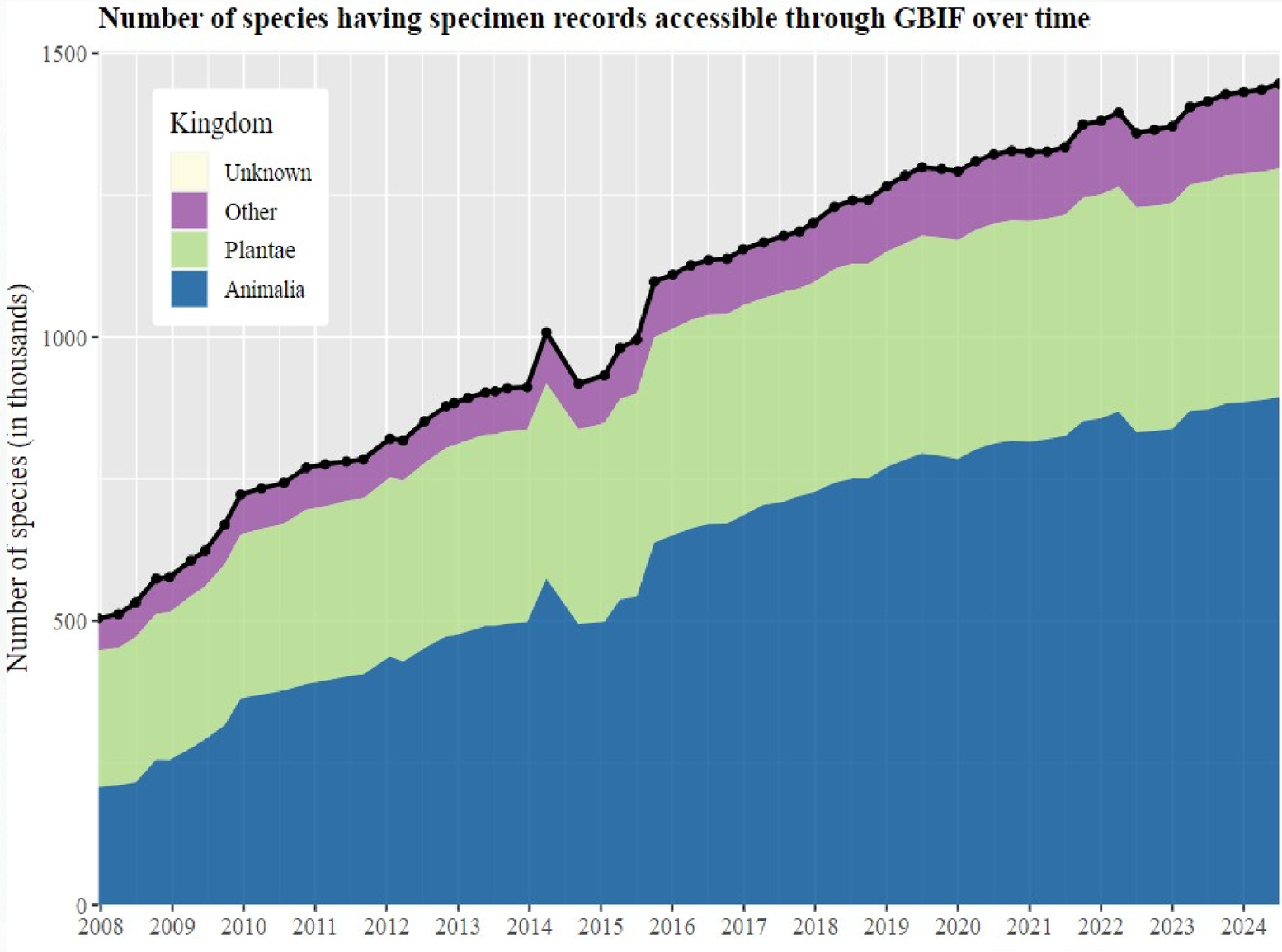
A brief talk about techniques I discovered this past month as well as some fascinating research in biodiversity classification

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Machine Learning for Species Classification

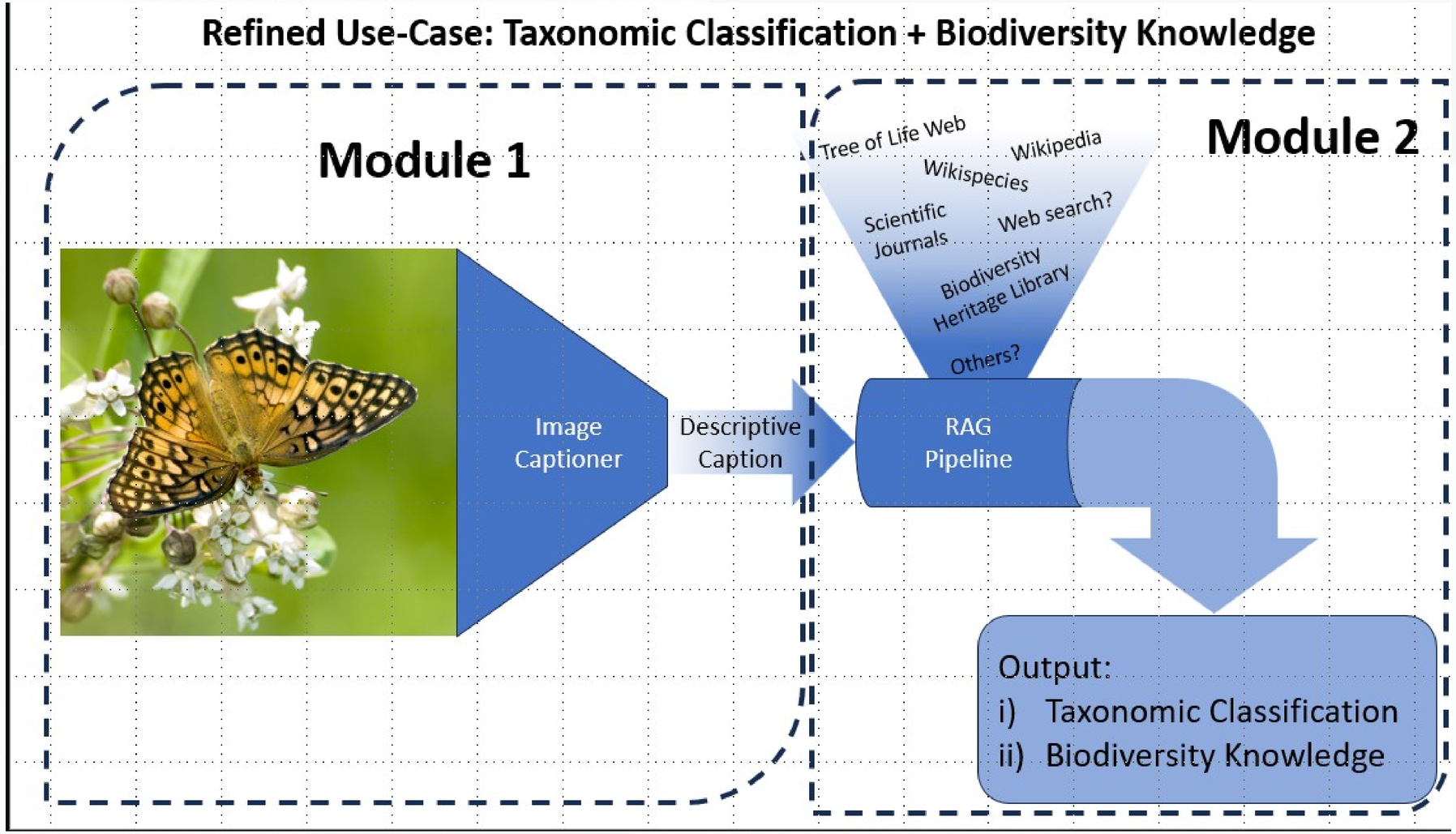
Up to one million species are threatened with extinction, many within decades. [1]

Monitoring Biodiversity has proven to be a challenge for many decades however machine learning has grown to be a big part of the developing solution thanks to increasingly available biodiversity data.



[1] https://www.un.org/en/climatechange/science/climate-issues/biodiversity
[2] https://www.gbif.org/analytics/global

Nate Lesperance's RAG Project



Refined Use-Case: Taxonomic Classification + Biodiversity Knowledge

True label: *Euptoieta claudia*



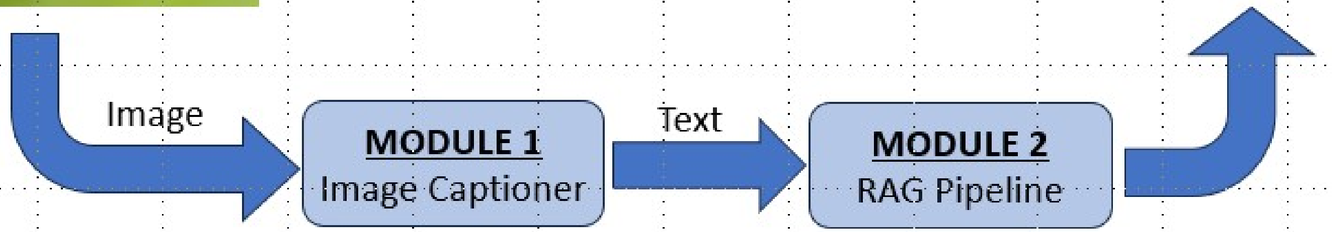
Result

Based on the information provided, the organism you most likely have in the image is a butterfly from the genus **Euptoieta**.

Although I don't have an image to visually confirm this, the description you've provided aligns with the common characteristics of the Euptoieta genus, particularly **Euptoieta hegesia**, also known as the variegated fritillary or the banded fritillary.

Within the context of the information you've provided, here's why Euptoieta is a suitable candidate:

1. Wing color and pattern: Euptoieta butterflies have distinctive brown and black wings with white or yellow bands, which matches the description of the wing color and pattern seen in your image.
2. **Floral preference:** Your description indicates that the butterfly was seen perching on a white flower, which is consistent with the floral preferences of Euptoieta species. They are often found on white or yellow flowers, especially those in the umbellifer family.
3. Range and habitat: Different Euptoieta species are found in various regions, and some are common in suburban areas



Datasets

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INATURALIST 2019

1010 Species
268,000 images
Crowd-Sourced Images

TREE OF LIFE 10M+

45,000 Species
10M+ images
Comprised of iNaturalist, BioScan-1M, Encyclopedia of Life (EoL)

INATURALIST 2021

10,000 Species
2.7M images
‘Mini’ has 500K images
Crowd-Sourced Images

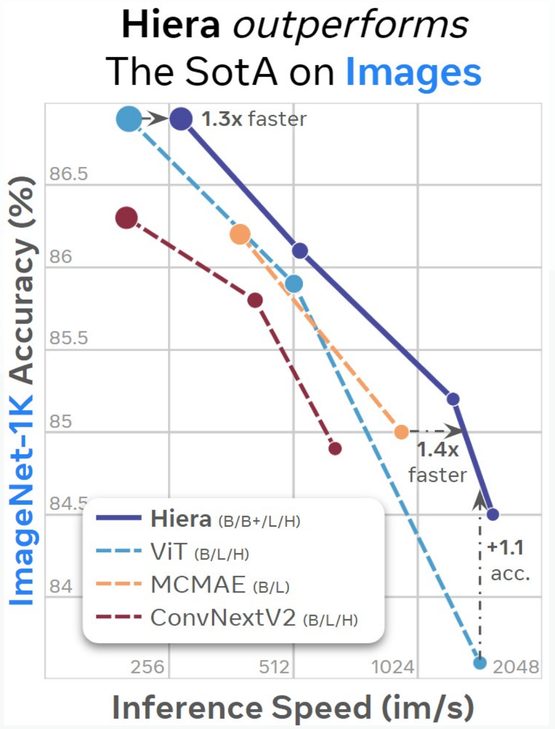
RARE SPECIES

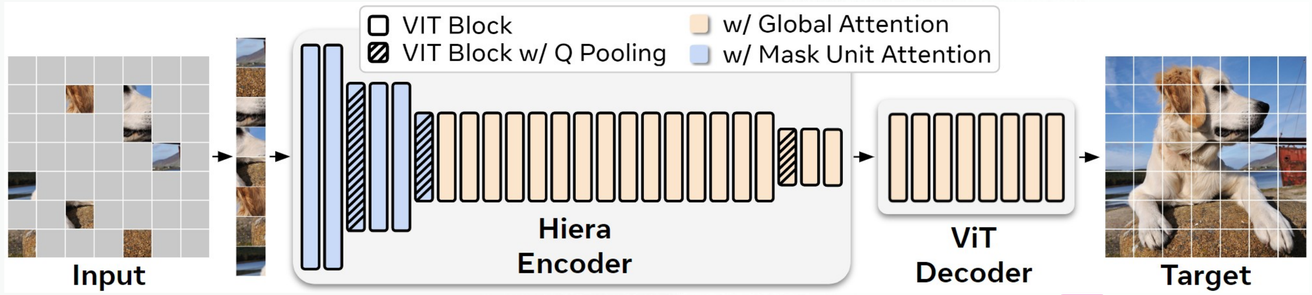
400 Species
12,000 images
Data from Encyclopedia of Life

Model 1: Hiera

"A Hierarchical Vision Transformer without the Bells-and-Whistles"

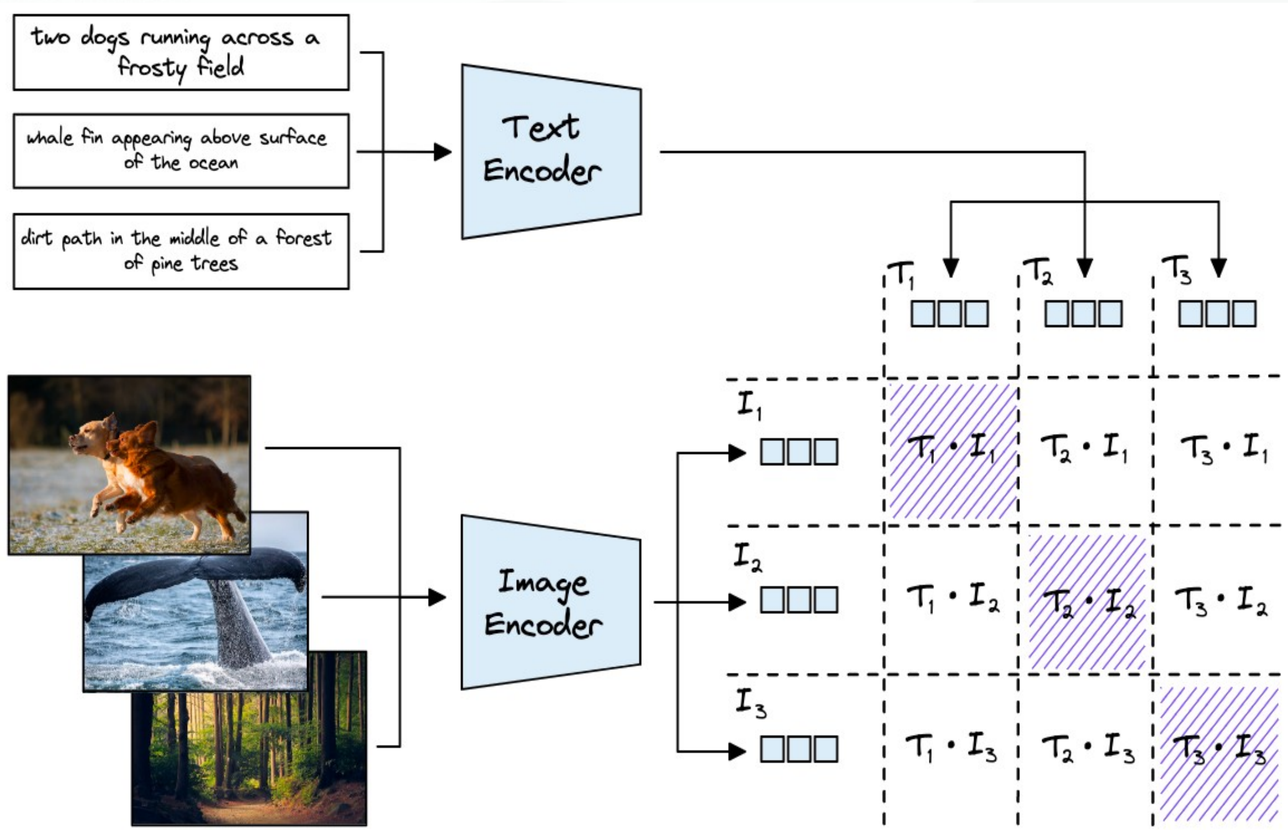
- Hiera is a hierarchical vision transformer that is fast, powerful, and, above all, simple. It outperforms the state-of-the-art across a wide array of image and video tasks while being much faster.
- Hierarchial Vision Transformers progressively reduce the spatial resolution of the input image while increasing the number of feature channels as the network goes deeper.
- Hiera uses **Masked Autoencoder (MAE) Training** to naturally learn spatial biases by reconstructing masked portions of the input image.



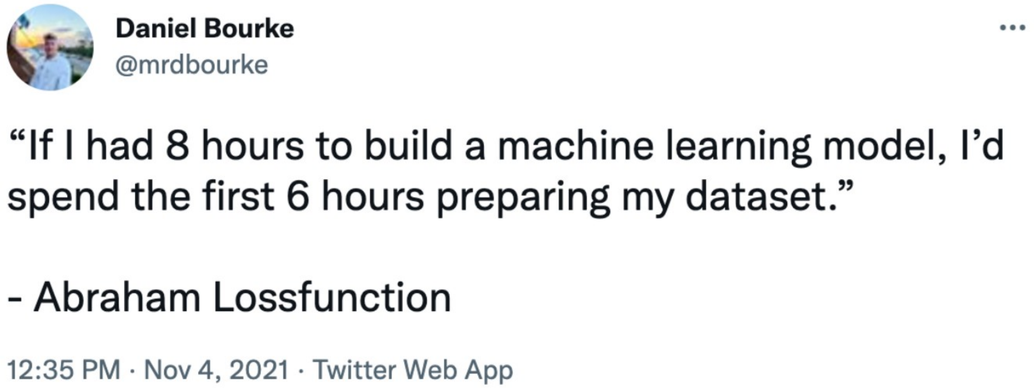


Model 2: CLIP

- CLIP (Contrastive Language-Image Pre-Training) is a neural network trained on a variety of (image, text) pairs.
- It uses Vision Transformers and Text Transformers to create separate embeddings.
- Both the image and text embeddings are projected into a shared embedding space where similar images and texts are close together. This shared space enables the model to compare images and text directly..
- Very effective for Zero-Shot Learning



The Training Process



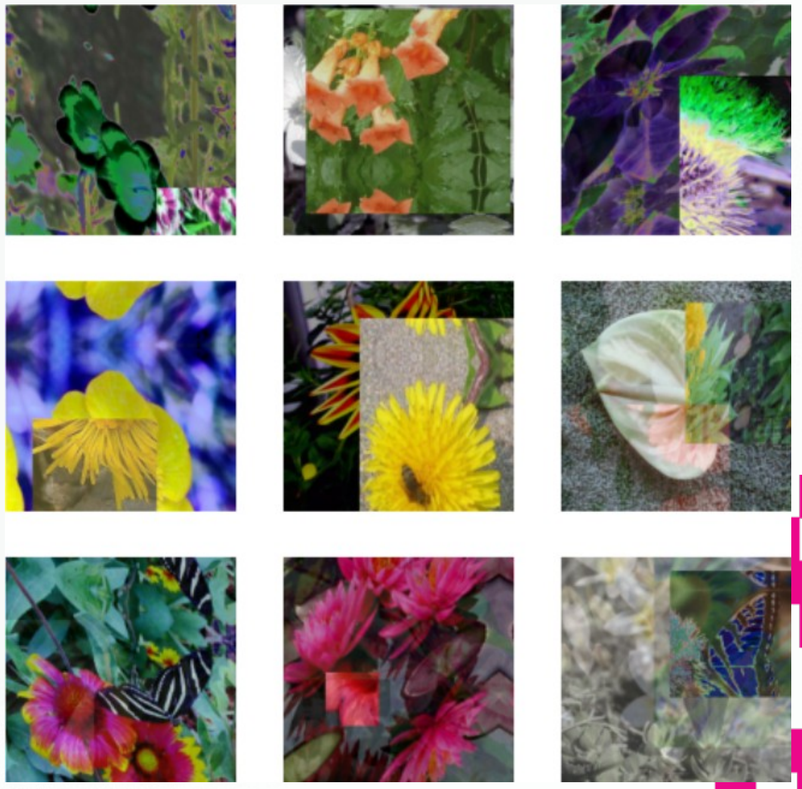
Data Augmentations

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After Random Augment



After RandAugment, Cutmix and Mixup



Additional Parameters

Label Smoothing: Prevents model becoming overconfident in it’s predictions

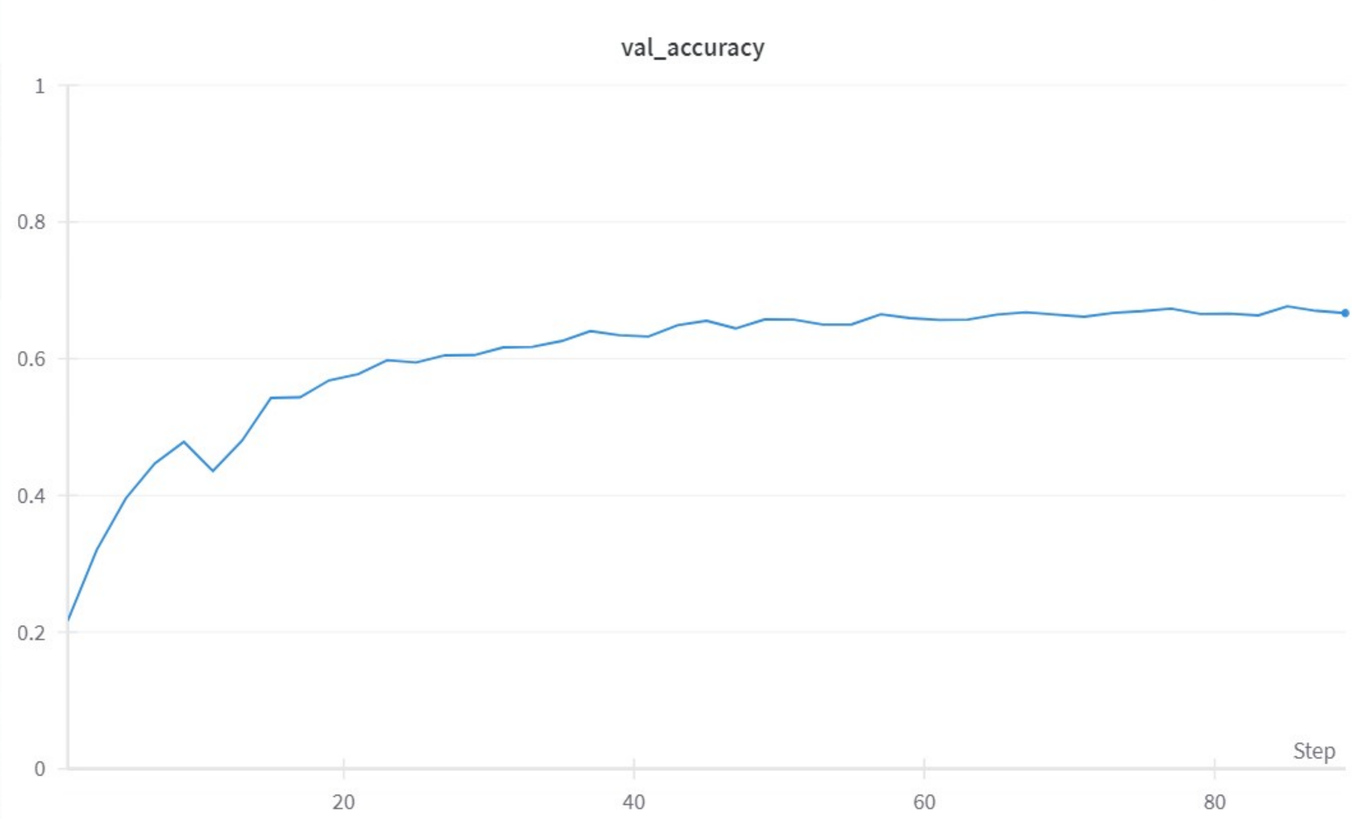
Cosine Decay LR Scheduler: This allows the model to take larger steps initially and smaller steps as it converges, leading to better fine-tuning.

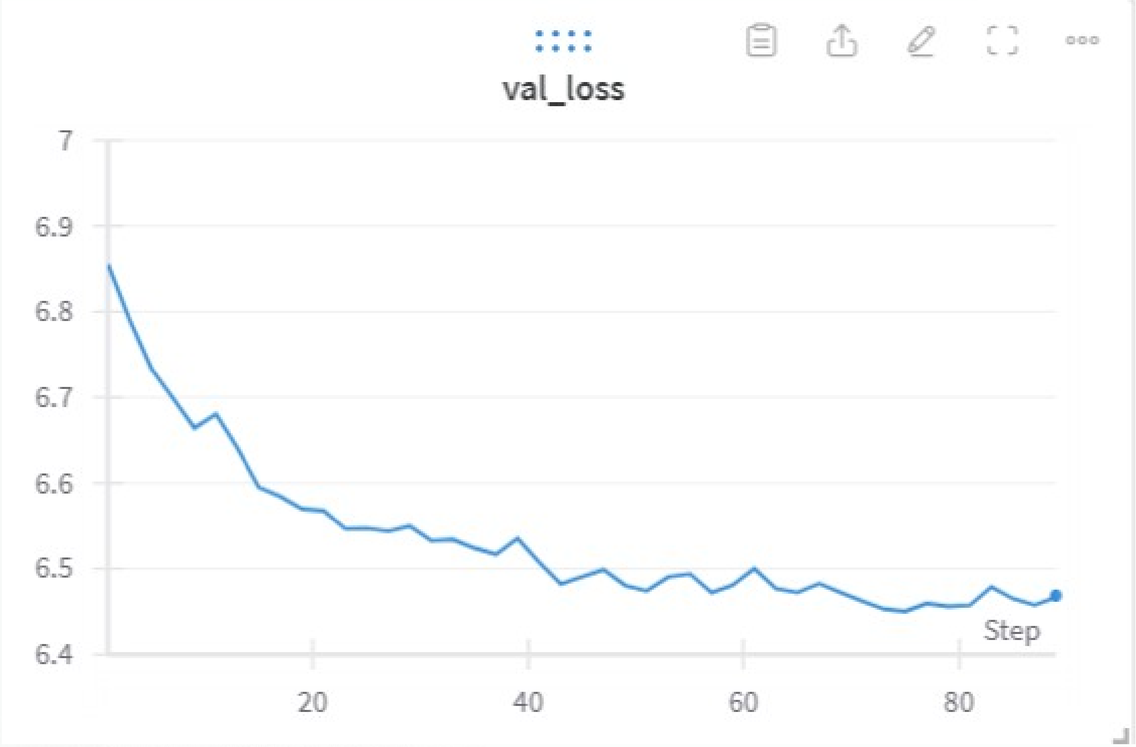
Drop Path and Layer Wise Decay: Drop Path helps avoid overfitting by randomly dropping parts of the model during training. Layer-wise decay ensures that lower layers, which capture more general features, are not overly modified, while upper layers, which learn species-specific features, receive more updates.

Results with Hiera(T) on iNaturalist 2019

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Accuracy peaks at **0.7**

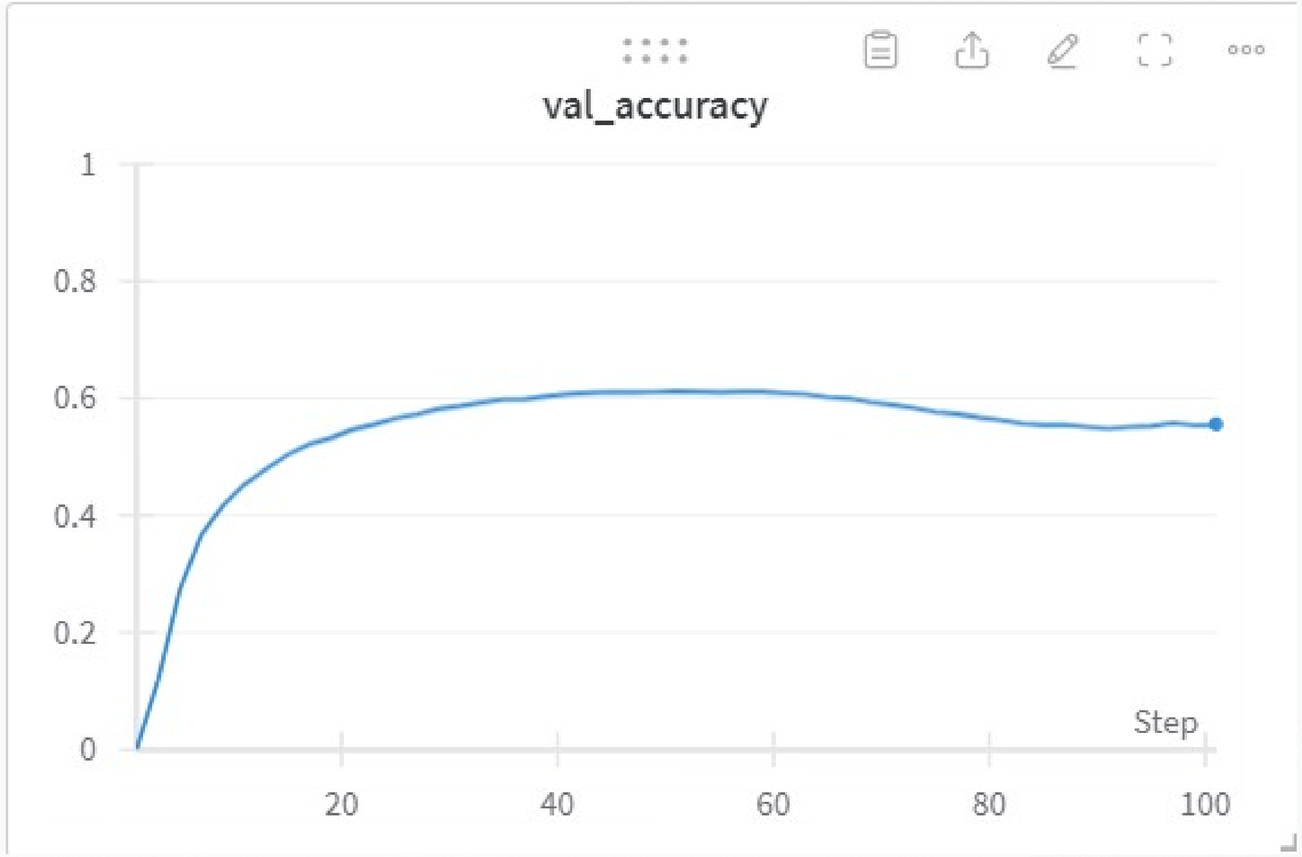


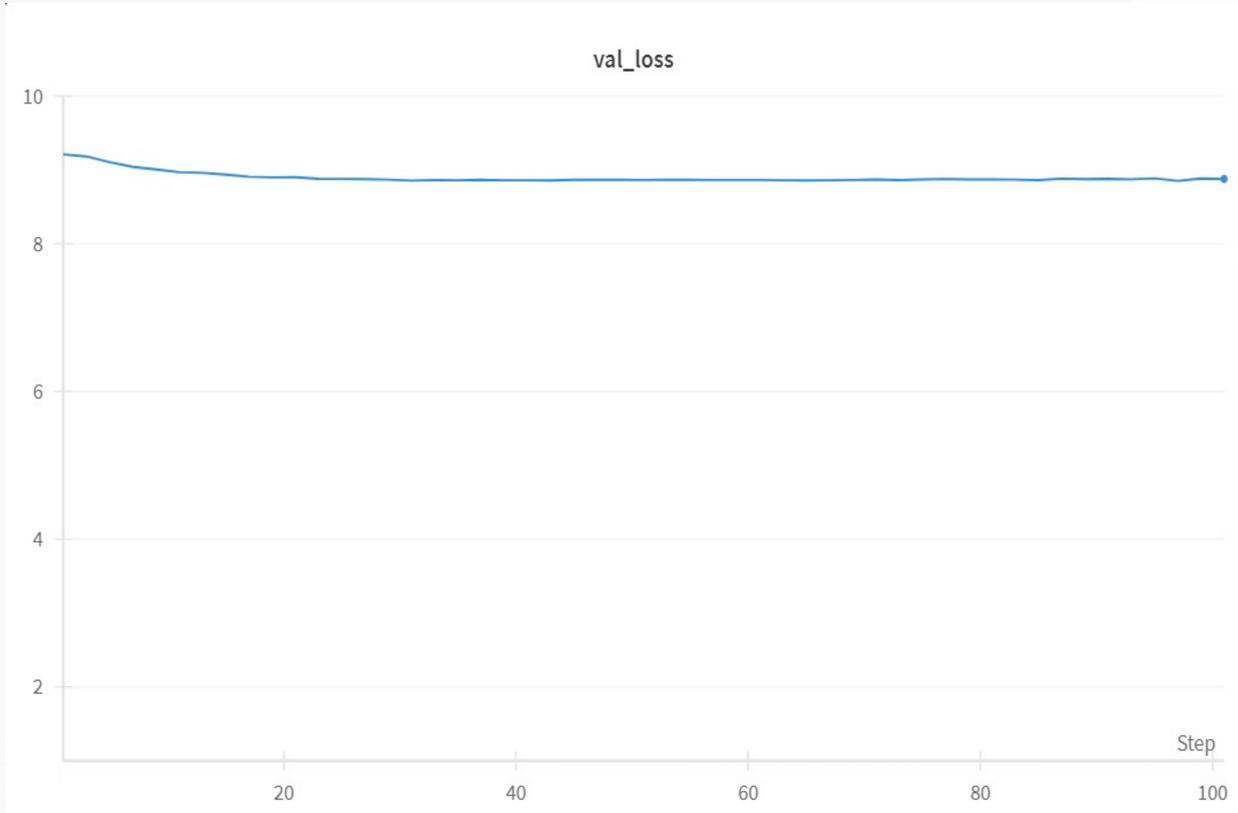


Results with Hiera(T) on iNaturalist 2021 (Mini)

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Accuracy peaked at **0.61**
(same as WaveMix-256/16)





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CLIP Evaluations

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Creating Clip Labels and Embeddings

```python
# Load categories
categories_path = '/scratch/ssd004/scratch/junejory/categories.json'
with open(categories_path, 'r') as f:
    categories = json.load(f)

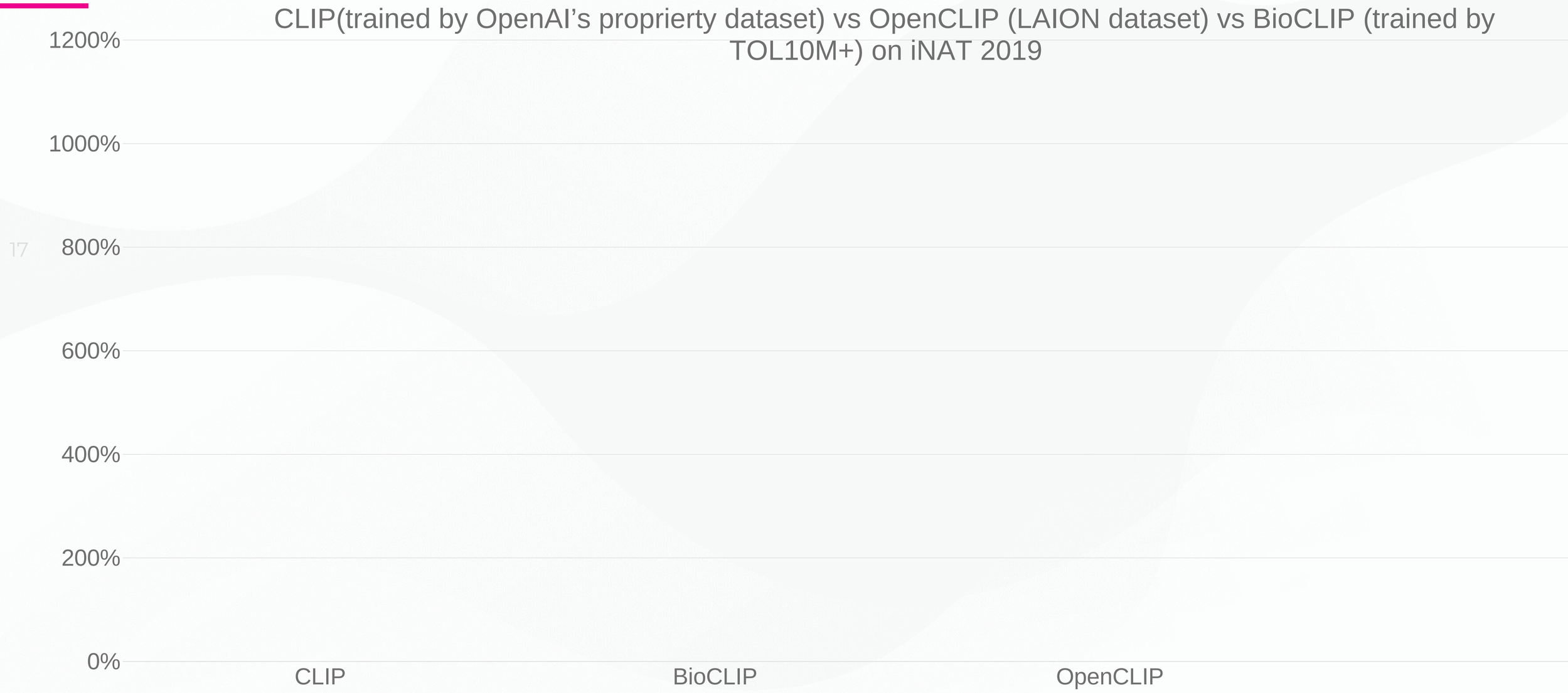
# Generate sentences for each category
clip_labels = [f"a photo of a {category['name']}" for category in categories]

# Load annotations
annotations_path = '/scratch/ssd004/scratch/junejory/val2019.json'
with open(annotations_path, 'r') as f:
    annotations = json.load(f)
```

```
clip_labels
```

```
['a photo of a Amanita vaginata',
 'a photo of a Amanita flavoconia',
 'a photo of a Amanita calyptroderma',
 'a photo of a Amanita jacksonii',
 'a photo of a Amanita phalloides',
 'a photo of a Amanita muscaria',
 'a photo of a Amanita bisporigera',
 'a photo of a Amanita gemmata',
 'a photo of a Amanita pantherina',
 'a photo of a Amanita velosa',
 'a photo of a Amanita augusta',
 'a photo of a Amanita rubescens',
 'a photo of a Cicindela aurulenta',
 'a photo of a Cicindela punctulata',
 'a photo of a Cicindela ocellata',
 'a photo of a Cicindela campestris',
 'a photo of a Cicindela oregona',
```

Battle of the CLIPS

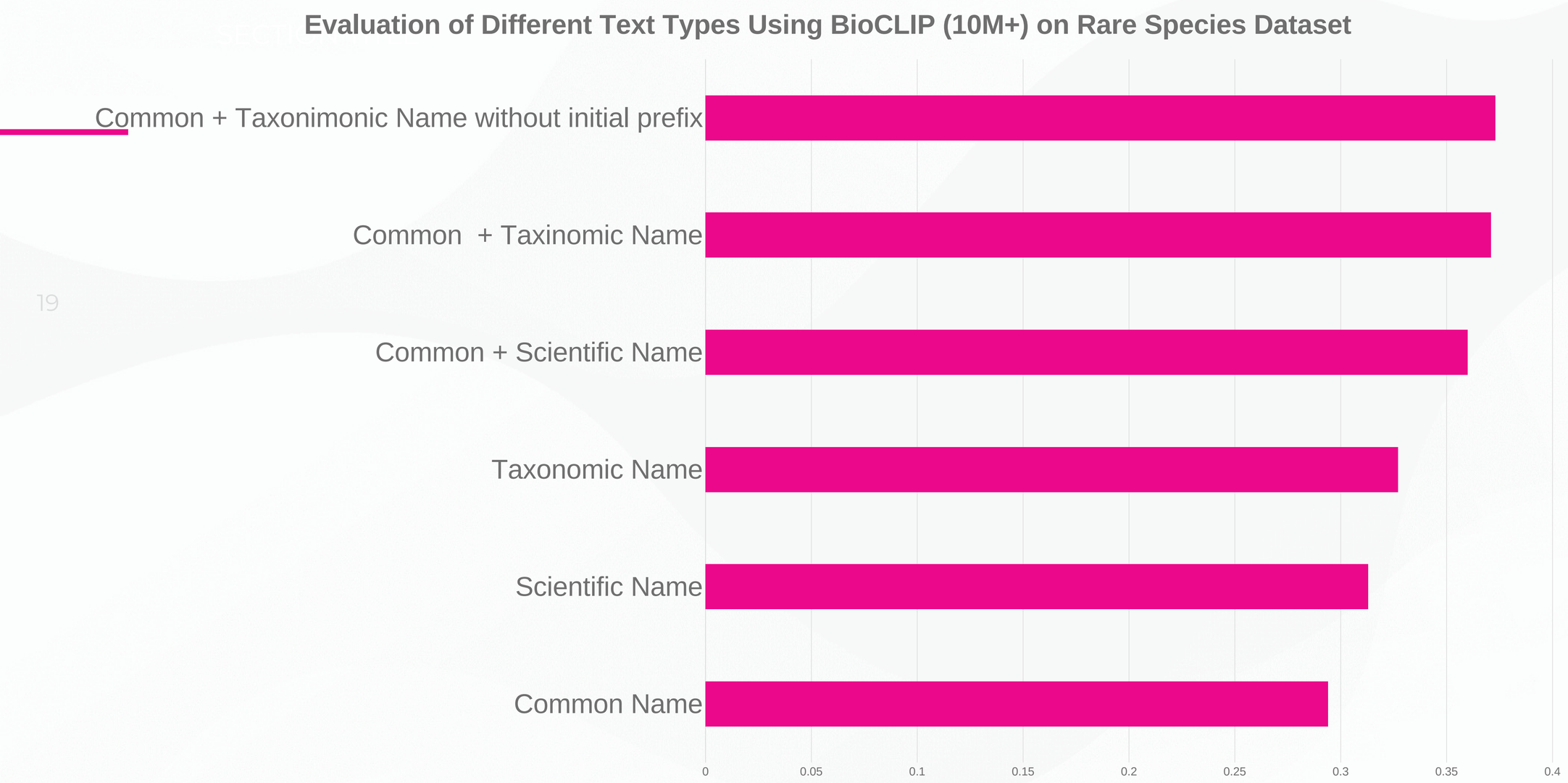


Text Types for BioCLIP

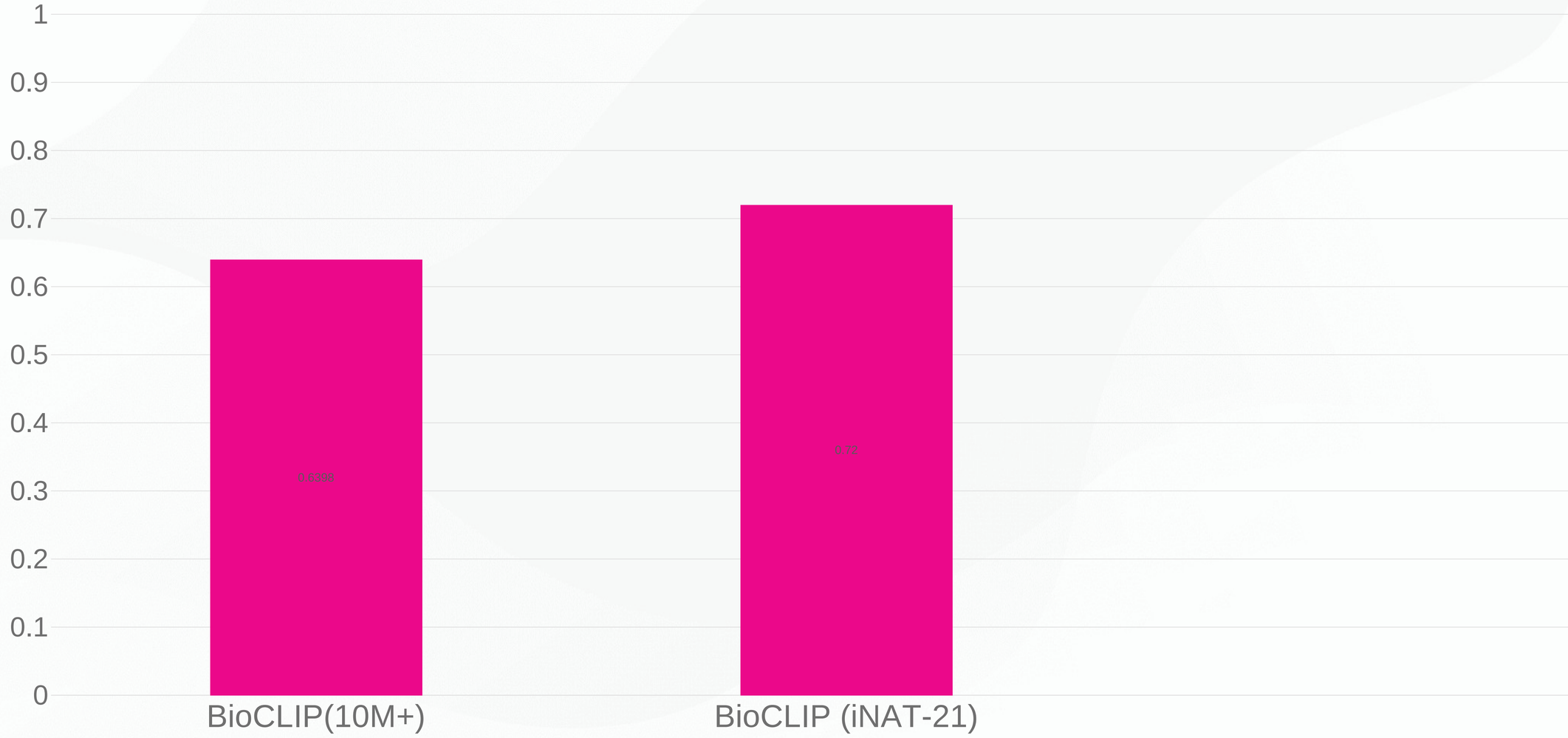
- Now that we've determined BioClip's efficiency, we wanted to see what types of text types for the CLIP Labels worked best for species classification.
- We used the Rare Species Dataset as an evaluation metric as BioCLIP considers these as previously unseen species in it's training data.
- Additionally, we wanted to see what the difference was in using a label such as "a photo of" as a prefix to text types.

Text Type	Example
Common	black-billed magpie
Scientific	*Pica hudsonia*
Taxonomic	*Animalia Chordata Aves Passeriformes Corvidae Pica hudsonia*
Scientific + Common	*Pica hudsonia* with common name black-billed magpie
Taxonomic + Common	*Animalia Chordata Aves Passeriformes Corvidae Pica hudsonia* with common name black-billed magpie

Battle of the Text Types



Additional Comparisons
BioCLIP(10M+) vs BioCLIP(iNat-21) on iNat-21



Challenges Faced

- Memory Considerations
- Storage Problems
- iNaturalist is highly imbalanced causing data offsets
- Hard to trace back problems when Epochs would take 30 minutes each

Lessons Learned

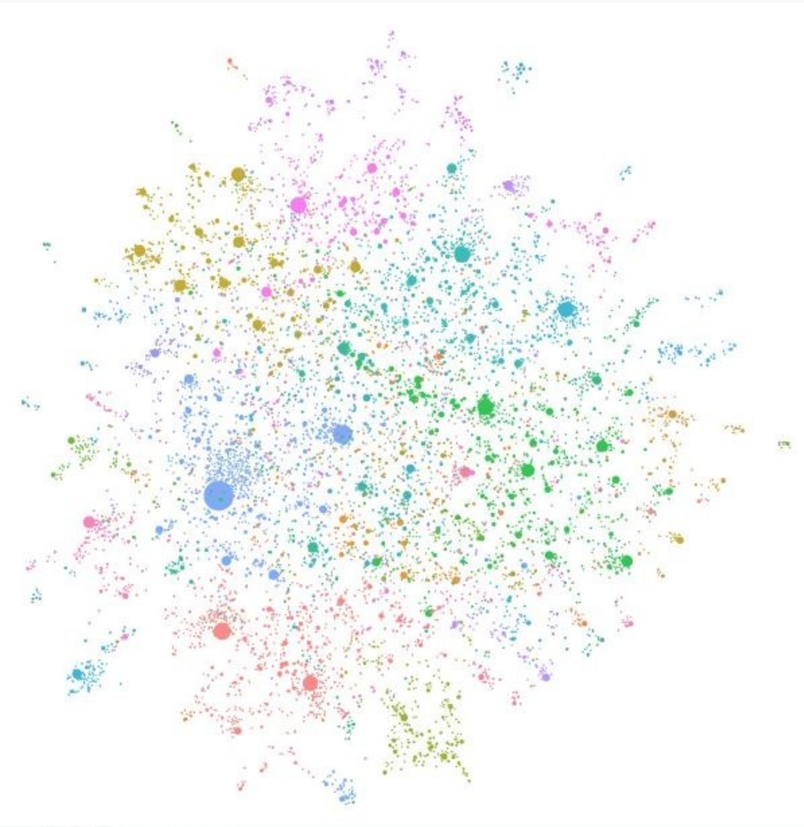
- Iterate Iterate Iterate!
- Get to know your Dataset, especially large ones like iNaturalist
- Large Datasets take TIME. Don’t be afraid to use more power (and batch size)

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Future RAG methologies and other Biodiversity Projects

Graph RAG

- Combines the strengths of **graph neural networks (GNNs)** and **retrieval-augmented generation** (RAG).
- Leverages **structured knowledge graphs** to enhance classification and inference tasks.
- Incorporates **retrieval mechanisms** to dynamically fetch relevant data during model training and inference.
- **Species Identification:** Improves species classification by integrating taxonomic hierarchies and ecological relationships.
- This may also aid the model in identifying different transitions of a species like a tadpole turning into a frog

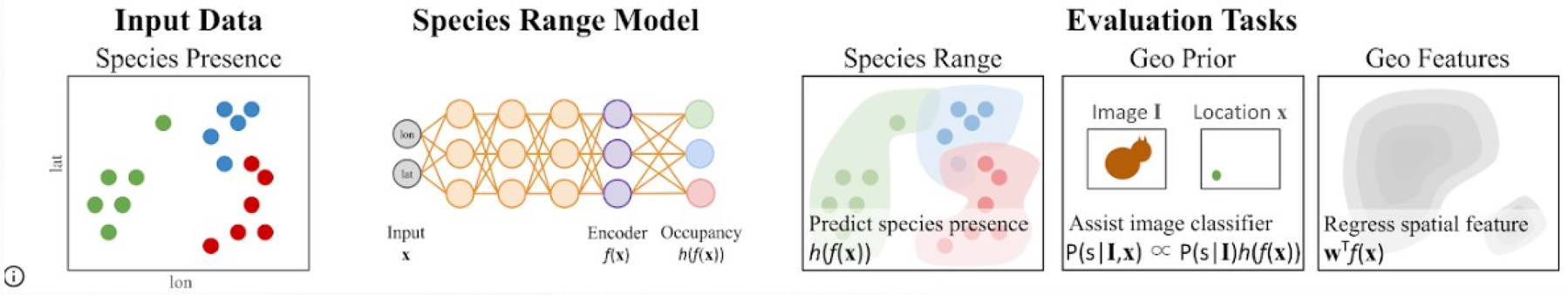


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Interesting Read that also features iNaturalist

- Spatial Implicit Neural Representations for Global-Scale Species Mapping by Elijah Cole
- Spatial Implicit Neural Representations (SINRs) are used to predict species presence/absence across locations by leveraging large-scale, crowdsourced datasets, enabling the simultaneous estimation of geographical ranges for 47,000 species.

- Scalability and Improved Predictions: The approach shows better performance as the number of species and data increases, efficiently handling noisy and biased data to approximate expert-developed range maps.



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Credits:
Thank you to Graham Taylor and the Vector Team for giving me this oppurtunity
Thank you to Nate for meeting with me every week to discuss the project!
Thank you to the MLRG team for having me!